

A Screening Method for Interaction Effects in SVDD with Mixed-type Features

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Introduction

RESEARCH BACKGROUND

Mixed-type data contain both continuous and categorical variables:

$$\mathbf{x}_i = \left(\mathbf{x}_i^{(c)}, \mathbf{x}_i^{(g)} \right),$$

where $\mathbf{x}_i^{(c)}$ denotes continuous variables and $\mathbf{x}_i^{(g)}$ denotes categorical variables.

In such data, an anomaly may arise not only from a single extreme variable, but also from an unusual combination of variables.

- ▶ Anomalies are usually rare and anomaly labels may be unavailable.
- ▶ SVDD is useful because it learns the normal region using normal observations only.
- ▶ However, interaction-driven anomalies may be difficult to detect using only original input variables.
- ▶ This thesis focuses on rare anomalies driven by mixed-type interactions.

TARGET: INTERACTION-DRIVEN ANOMALIES

- ▶ This thesis focuses on rare anomalies caused by interaction structures in mixed-type data.
- ▶ In particular, the target anomaly is not defined only by an extreme value of a single variable. Instead, it is generated by an unusual combination of variables.

Target interaction structures

Interaction type	Meaning
Continuous–categorical	A continuous variable and a categorical variable jointly form an abnormal pattern.
Categorical–categorical	A combination of categorical levels forms an abnormal pattern.

Using the notation of this thesis, the screening targets are

$$\mathcal{C} \times \mathcal{G} = \{(c, g) : c \in \mathcal{C}, g \in \mathcal{G}\}, \quad \mathcal{G} \times \mathcal{G} = \{(g, h) : g, h \in \mathcal{G}, g < h\}$$

Goal: detect rare anomalies arising from $\mathcal{C} \times \mathcal{G}$ or $\mathcal{G} \times \mathcal{G}$ structures.

LIMITATION OF ORIGINAL SVDD

Original SVDD uses the given input variables.

- ▶ If interaction terms are not explicitly represented, SVDD may not fully capture interaction-driven anomaly structures.
- ▶ A simple solution is to add all possible second-order interaction terms.
- ▶ However, the number of candidate interactions rapidly increases:

$$\#\{\text{candidate interactions}\} = O(p^2)$$

Thus, interaction screening is needed before adding interaction features.

RFE PSEUDO-SAMPLES (SANZ ET AL., 2018)

Pseudo-sample construction

- ▶ $X_1, \dots, X_p$: standardized features
- ▶ Fix a target feature $X_j, j = 1, \dots, p$.
- ▶ Choose quantile levels $q_r \in (0, 1)$, and define

$$z_{jr} = Q(X_j; q_r), \quad r = 1, \dots, R,$$

where $Q(\cdot; q)$ is the empirical quantile operator.

- ▶ Pseudo-sample matrix for X_j :

$$S_j = \begin{bmatrix} z_{j1} & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ z_{jR} & 0 & \cdots & 0 \end{bmatrix}, \quad \mathbf{s}_{j,r} = (0, \dots, 0, \underbrace{z_{jr}}_{j\text{-th}}, 0, \dots, 0)^\top \in \mathbb{R}^p$$

is a pseudo-sample vector that varies only in the j -th coordinate.

- ▶ All other variables are fixed at 0, so only the marginal effect of X_j is explored across the pseudo-samples.

Proposed Method

STAGE-1 SVDD AND CENTER DISTANCE

Stage-1 SVDD is fitted using only normal training observations:

$$\mathcal{X}_0 = \{\mathbf{x}_1, \dots, \mathbf{x}_{n_0}\}$$

In the feature space, the SVDD center is

$$\mathbf{a}_\phi = \sum_{i=1}^{n_0} \alpha_i \phi(\mathbf{x}_i)$$

For any point $\mathbf{z}$, define the center-distance function:

$$\Delta(\mathbf{z}) = \|\phi(\mathbf{z}) - \mathbf{a}_\phi\|^2$$

Using kernels,

$$\Delta(\mathbf{z}) = K(\mathbf{z}, \mathbf{z}) - 2 \sum_{i=1}^{n_0} \alpha_i K(\mathbf{z}, \mathbf{x}_i) + \sum_{i=1}^{n_0} \sum_{j=1}^{n_0} \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j)$$

Why use $\Delta(\cdot)$ for screening?

Stage-1 SVDD learns the center of the normal region. If pseudo-samples generated from a candidate interaction are far from this center, then the interaction may form an abnormal mixed-type pattern. Therefore, the average value of $\Delta(\mathbf{s})$ is used as a label-free interaction score.

STAGE-1 MIXED-KERNEL SVDD

Each observation is decomposed as

$$\mathbf{x}_i = \left(\tilde{\mathbf{x}}_i^{(c)}, \mathbf{x}_i^{(g)} \right)$$

Continuous variables are min-max scaled, while categorical variables keep their category levels.

The mixed kernel is

$$K_{\text{mix}}(\mathbf{x}_i, \mathbf{x}_j) = \exp \left[\tau \left\{ \text{CK}(\mathbf{x}_i^{(g)}, \mathbf{x}_j^{(g)}) - \|\tilde{\mathbf{x}}_i^{(c)} - \tilde{\mathbf{x}}_j^{(c)}\|^2 \right\} \right], \quad \tau > 0$$

- ▶ $\text{CK}(\cdot, \cdot)$: Categorical kernel
- ▶ $\|\tilde{\mathbf{x}}_i^{(c)} - \tilde{\mathbf{x}}_j^{(c)}\|^2$: continuous distance term (Gaussian kernel)
- ▶ K_{mix} combines both types of information in one kernel.

TUNING PARAMETER SELECTION

The mixed kernel contains the scale parameter τ .

Issue: in one-class learning, anomaly labels are unavailable.

⇒ supervised cross-validation using recall or F1-score cannot be used.

Therefore, τ is selected using the SVDD target error estimate (Tax and Duin, 2004):

$$\tilde{E}_{\text{LOO}} \leq \frac{\#SV + \#errors}{n_0}$$

For a candidate grid $\mathcal{T}$,

$$\tau^* = \min \left\{ \tau \in \mathcal{T} : \frac{\#SV(\tau) + \#errors(\tau)}{n_0} \geq \epsilon \right\}$$

In the simulation study,

$$\epsilon = 0.05$$

PSEUDO-SAMPLE CONSTRUCTION

This thesis screens two types of second-order interactions:

$$\mathcal{C} \times \mathcal{G} = \{(c, g) : c \in \mathcal{C}, g \in \mathcal{G}\}, \quad \mathcal{G} \times \mathcal{G} = \{(g, h) : g, h \in \mathcal{G}, g < h\}$$

A baseline vector is constructed from normal training data:

$$\mathbf{x}^{\text{base}} = ((\tilde{m}_c)_{c \in \mathcal{C}}, (b_g^*)_{g \in \mathcal{G}})^\top,$$

where

$$\tilde{m}_c = \text{median}(\tilde{X}_c), \quad b_g^* = \text{mode}(X_g)$$

Pseudo-sample idea

For each candidate interaction, only the variables in the candidate pair are varied. All other variables are fixed at the baseline vector.

- ▶ $\mathcal{C} \times \mathcal{G}$: use conditional quantiles of $\tilde{X}_c$ given $X_g = \ell$
- ▶ $\mathcal{G} \times \mathcal{G}$: use representative joint cells of (X_g, X_h)

INTERACTION SCORE

For each candidate interaction, pseudo-samples are evaluated by the Stage-1 center-distance function $\Delta(\cdot)$.

C×G interaction score

$$D_{c,g} = \frac{1}{T_{c,g}} \sum_{s \in S_{c,g}} \Delta(s)$$

G×G interaction score

$$D_{g,h} = \frac{1}{T_{g,h}} \sum_{s \in S_{g,h}} \Delta(s)$$

Interpretation

A larger score means that pseudo-samples generated from the candidate interaction are farther from the Stage-1 SVDD center on average. Thus, the interaction is more likely to represent a pattern that deviates from the normal region.

INTERACTION RANKING AND SELECTION

All candidate interactions are ranked by their center-distance score:

$$D_{c,g} \quad \text{or} \quad D_{g,h}$$

The number of selected interactions is defined as

$$k = \frac{n_0}{\log n_0}$$

Reason for using $k = n_0 / \log n_0$:

This cutoff is not claimed to be theoretically optimal. It is a practical screening cutoff motivated by the screening literature (Fan and Lv, 2008; Li et al., 2012; Yang et al., 2019). The form $n / \log n$ increases with sample size but grows more slowly than n . Since Stage-2 SVDD is trained using n_0 normal observations, we use $n_0 / \log n_0$ in this thesis.

STAGE-2 SVDD WITH SELECTED INTERACTIONS

In the mixed interaction setting, $C \times G$ and $G \times G$ candidates are combined:

$$\mathcal{H}_{\text{all}} = \{((c, g), D_{c,g}) : (c, g) \in \mathcal{C} \times \mathcal{G}\} \cup \{((g, h), D_{g,h}) : (g, h) \in \mathcal{G} \times \mathcal{G}, g < h\}.$$

The selected interaction set is

$$I_{\text{all}} = \text{Top}_k(\mathcal{H}_{\text{all}}).$$

Selected interactions are added as new numerical interaction features:

$$\psi_{c,g}(\mathbf{x}_i) = \tilde{x}_{ic}x_{ig}, \quad \psi_{g,h}(\mathbf{x}_i) = x_{ig}x_{ih}.$$

The expanded input vector is

$$\mathbf{x}_i^* = \left(\tilde{\mathbf{x}}_i^{(c)}, \mathbf{x}_i^{(g)}, \{\psi_{c,g}(\mathbf{x}_i)\}, \{\psi_{g,h}(\mathbf{x}_i)\} \right)$$

Stage-2 SVDD is refitted using only normal observations, and τ is reselected on the expanded feature space.

Simulation Study

SIMULATION STUDY: PURPOSE

The simulation study is designed to generate rare anomalies driven by mixed-type interaction structures.

Main purpose

- ▶ To evaluate whether PIS-SVDD can detect rare interaction-driven anomalies.
- ▶ To compare PIS-SVDD with SVDD(no screening).
- ▶ To examine whether the proposed screening step remains useful across linear, quadratic, and sine-based nonlinear structures.

Simulation focus

rare anomaly + mixed-type interaction + various generating structures

COMMON SIMULATION SETTING

Component	Setting
Training sample size	$n_{\text{train}} = 1000$
Training normal / anomaly	$n_0^{\text{train}} = 950, n_1^{\text{train}} = 50$
Test sample size	$n_{\text{test}} = 2000$
Test normal / anomaly	$n_0^{\text{test}} = 1900, n_1^{\text{test}} = 100$
Anomaly proportion	$\pi_{\text{out}} = 0.05$
Repetitions	50
Total number of variables	$p = 100$
Main variables	$X_1, \dots, X_6$
Noise variables	$X_7, \dots, X_{100}$

DATA GENERATION: NORMAL VS ANOMALY DISTRIBUTIONS

Normal and anomaly observations are generated from different distributions:

$$\mathbf{x}_i \mid Y_i = 0 \sim F_0, \quad \mathbf{x}_i \mid Y_i = 1 \sim F_1$$

For the main continuous variables,

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \bigg| Y = 0 \sim N_2 \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix} \right],$$

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \bigg| Y = 1 \sim N_2 \left[\begin{pmatrix} 1.25 \\ 1.15 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix} \right]$$

For the main categorical variables used in the target interactions, $X_3, \dots, X_6$ are generated from Bernoulli distributions.

The remaining variables are independent noise variables:

$$X_7, \dots, X_{100} \stackrel{\text{ind}}{\sim} N(0, 1)$$

LATENT SCORE AND ANOMALY PROBABILITY

For each simulation setting, the latent score η_i is designed to contain weak main effects and relatively strong interaction effects.

$$\eta_i = \log \frac{p_i}{1 - p_i}, \quad p_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

Given the generated p_i , the anomaly label is generated by

$$Y_i \sim \text{Bernoulli}(p_i), \quad Y_i \in \{0, 1\}$$

Simulation design

Normal observations are generated from F_0 , while anomaly observations are generated from F_1 . The anomaly proportion is fixed at 5% in both training and test data.

Thus, observations with stronger interaction structures have larger η_i and p_i , and are more likely to be generated as anomalies.

SIMULATION SCENARIOS

Three cases are considered according to the complexity of the anomaly-generating structure.

Case	Structure	Purpose
Case 1	Linear	Basic interaction-driven anomaly
Case 2	Quadratic nonlinear	Nonlinear main effects + interactions
Case 3	Sine-based nonlinear	More complex nonlinear interaction pattern

For each case, three interaction configurations are considered.

Configuration	Relevant interaction type
$C \times G$	Two continuous–categorical interactions
$G \times G$	Two categorical–categorical interactions
Mixed	One $C \times G$ interaction and one $G \times G$ interaction

CASE 1: LINEAR STRUCTURE

Case 1 uses a linear latent score:

$$\eta_i = \beta_0 + \underbrace{\text{weak main effects}}_{\beta_{\text{main}}} + \underbrace{\text{strong interaction effects}}_{\beta_{\text{int}}}$$

Setting	Target interaction terms	Interaction coefficients
C×G	$x_{i1}x_{i3}, x_{i2}x_{i4}$	$\beta_5 = \beta_6 = 4.80$
G×G	$x_{i3}x_{i4}, x_{i5}x_{i6}$	$\beta_7 = \beta_8 = 4.20$
Mixed	$x_{i1}x_{i3}, x_{i4}x_{i5}$	$\beta_7 = 7.20, \beta_8 = 6.50$

CASE 1: EXAMPLE LOGIT FORMS

C×G setting

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \underbrace{\beta_5 x_{i1} x_{i3} + \beta_6 x_{i2} x_{i4}}_{\text{C} \times \text{G interactions}}$$

G×G setting

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \beta_6 x_{i6} + \underbrace{\beta_7 x_{i3} x_{i4} + \beta_8 x_{i5} x_{i6}}_{\text{G} \times \text{G interactions}}$$

Mixed setting

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \underbrace{\beta_6 x_{i1} x_{i3}}_{\text{C} \times \text{G}} + \underbrace{\beta_7 x_{i4} x_{i5}}_{\text{G} \times \text{G}}$$

CASE 2: QUADRATIC NONLINEAR STRUCTURE

Case 2 includes quadratic terms in the latent score:

$$\eta_i = \beta_0 + \underbrace{\text{linear and quadratic main effects}}_{\text{nonlinear marginal structure}} + \underbrace{\text{strong interaction effects}}_{\text{target anomaly structure}}$$

Setting	Target interaction terms	Interaction coefficients
C×G	$x_{i1}x_{i3}, x_{i2}x_{i4}$	$\beta_9 = \beta_{10} = 2.20$
G×G	$x_{i3}x_{i4}, x_{i5}x_{i6}$	$\beta_{13} = \beta_{14} = 1.50$
Mixed	$x_{i1}x_{i3}, x_{i4}x_{i5}$	$\beta_{11} = 8.50, \beta_{12} = 6.20$

CASE 2: COMPACT LOGIT FORMS

C×G setting

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \beta_3 x_{i2} + \beta_4 x_{i2}^2 + \beta_5 x_{i3} + \beta_6 x_{i3}^2 + \beta_7 x_{i4} + \beta_8 x_{i4}^2 \\ + \underbrace{\beta_9 x_{i1} x_{i3} + \beta_{10} x_{i2} x_{i4}}_{\text{C} \times \text{G interactions}}$$

G×G setting

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \cdots + \beta_{11} x_{i6} + \beta_{12} x_{i6}^2 \\ + \underbrace{\beta_{13} x_{i3} x_{i4} + \beta_{14} x_{i5} x_{i6}}_{\text{G} \times \text{G interactions}}$$

Mixed setting

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \cdots + \beta_{10} x_{i5}^2 + \underbrace{\beta_{11} x_{i1} x_{i3}}_{\text{C} \times \text{G}} + \underbrace{\beta_{12} x_{i4} x_{i5}}_{\text{G} \times \text{G}}$$

CASE 3: SINE-BASED NONLINEAR STRUCTURE

Case 3 uses sine functions to generate nonlinear interaction-driven anomalies:

$$\eta_i = \beta_0 + \sin \left(\underbrace{\text{weak main effects}}_{\text{small coefficients}} + \underbrace{\text{strong interaction effects}}_{\text{target structure}} \right)$$

Setting	Target interaction terms	Interaction coefficients
C×G	$x_{i1}x_{i3}, x_{i2}x_{i4}$	$\beta_5 = \beta_6 = 9.00$
G×G	$x_{i3}x_{i4}, x_{i5}x_{i6}$	$\beta_7 = \beta_8 = 1.60$
Mixed	$x_{i1}x_{i3}, x_{i4}x_{i5}$	$\beta_6 = 1.15, \beta_7 = 1.10$

CASE 3: COMPACT LOGIT FORMS

C×G setting

$$\eta_i = \beta_0 + \sin \left(\beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \underbrace{\beta_5 x_{i1} x_{i3} + \beta_6 x_{i2} x_{i4}}_{\text{C} \times \text{G interactions}} \right)$$

G×G setting

$$\eta_i = \beta_0 + \sin \left(\beta_1 x_{i1} + \cdots + \beta_6 x_{i6} + \underbrace{\beta_7 x_{i3} x_{i4} + \beta_8 x_{i5} x_{i6}}_{\text{G} \times \text{G interactions}} \right)$$

Mixed setting

$$\eta_i = \beta_0 + \sin \left(\beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \underbrace{\beta_6 x_{i1} x_{i3}}_{\text{C} \times \text{G}} + \underbrace{\beta_7 x_{i4} x_{i5}}_{\text{G} \times \text{G}} \right)$$

COMPARATIVE MODELS

Model	Role
PIS-SVDD	Proposed two-stage SVDD with interaction screening
SVDD(no screening)	Baseline one-class SVDD using original variables only
LASSO logistic regression	Supervised model with all two-way interactions
XGBoost	Supervised nonlinear tree-based model
Random Forest	Supervised ensemble tree-based model

- ▶ SVDD-based models use only normal observations for training.
- ▶ Supervised models use anomaly labels in the training data.

EVALUATION METRICS

Metric	Definition
Recall	$\frac{TP}{TP + FN}$
Precision	$\frac{TP}{TP + FP}$
F1-score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

- ▶ Anomalies are treated as the positive class.
- ▶ The anomaly proportion is only 5%.

Simulation Results

CASE 1 RESULTS: LINEAR

Structure	Model	TP	FP	TN	FN	Recall	F1
C×G 2	PIS-SVDD	100.000	45.600	1854.400	0.000	1.000	0.814
	SVDD	6.400	1.200	1898.800	93.600	0.064	0.115
	LASSO	16.000	84.600	1815.400	84.000	0.160	0.159
	XGB	51.170	71.480	1828.520	48.830	0.511	0.459
	RF	53.600	56.000	1844.000	46.400	0.536	0.509
G×G 2	PIS-SVDD	88.800	62.300	1837.700	11.200	0.888	0.707
	SVDD	4.400	70.800	1829.200	95.600	0.044	0.050
	LASSO	17.200	101.800	1798.200	82.800	0.172	0.156
	XGB	33.200	61.200	1838.800	66.800	0.332	0.338
	RF	27.000	66.000	1834.000	73.000	0.270	0.280
Mixed	PIS-SVDD	95.000	38.800	1861.200	5.000	0.950	0.812
	SVDD	4.200	10.600	1889.400	95.800	0.042	0.073
	LASSO	37.400	23.070	1876.930	62.600	0.374	0.466
	XGB	47.200	0.760	1899.240	52.800	0.472	0.638
	RF	21.400	0.400	1899.600	78.600	0.214	0.348

Values are averages over 50 repetitions.

CASE 1: INTERPRETATION

- ▶ In the $C \times G$ 2 setting, PIS-SVDD detected all anomalies:

$$\text{Recall} = 1.000, \quad \text{F1-score} = 0.814$$

- ▶ In the $G \times G$ 2 setting, PIS-SVDD achieved

$$\text{Recall} = 0.888, \quad \text{F1-score} = 0.707$$

- ▶ In the mixed setting, PIS-SVDD achieved

$$\text{Recall} = 0.950, \quad \text{F1-score} = 0.812$$

- ▶ SVDD(no screening) showed low recall and F1-score in all three settings.

CASE 2 RESULTS: QUADRATIC NONLINEAR

Structure	Model	TP	FP	TN	FN	Recall	F1
C×G 2	PIS-SVDD	91.400	64.200	1835.800	8.600	0.914	0.715
	SVDD	5.800	74.800	1825.200	94.200	0.058	0.064
	LASSO	38.500	90.520	1809.480	61.500	0.385	0.336
	XGB	54.700	81.624	1818.376	45.300	0.547	0.462
	RF	49.600	68.200	1831.800	50.400	0.496	0.455
G×G 2	PIS-SVDD	98.800	70.400	1829.600	1.200	0.988	0.734
	SVDD	3.800	69.200	1830.800	96.200	0.038	0.044
	LASSO	19.600	157.000	1743.000	80.400	0.196	0.141
	XGB	61.040	57.200	1842.800	38.960	0.610	0.559
	RF	52.200	85.200	1814.800	47.800	0.522	0.440
Mixed	PIS-SVDD	98.600	51.800	1848.200	1.400	0.986	0.787
	SVDD	4.200	10.600	1889.400	95.800	0.042	0.073
	LASSO	42.400	59.000	1841.000	57.600	0.424	0.423
	XGB	30.800	58.000	1842.000	69.200	0.308	0.326
	RF	33.200	77.400	1822.600	66.800	0.332	0.314

Values are averages over 50 repetitions.

CASE 2: INTERPRETATION

- ▶ In the $C \times G$ 2 setting, PIS-SVDD showed the highest performance:

$$\text{Recall} = 0.914, \quad \text{F1-score} = 0.715$$

- ▶ In the $G \times G$ 2 setting, PIS-SVDD achieved

$$\text{Recall} = 0.988, \quad \text{F1-score} = 0.734$$

- ▶ In the mixed setting, PIS-SVDD achieved

$$\text{Recall} = 0.986, \quad \text{F1-score} = 0.787$$

- ▶ SVDD(no screening) showed lower recall and F1-score in all three settings.

CASE 3 RESULTS: SINE-BASED NONLINEAR

Structure	Model	TP	FP	TN	FN	Recall	F1
C×G 2	PIS-SVDD	80.200	44.600	1855.400	19.800	0.802	0.714
	SVDD	7.000	93.000	1807.000	93.000	0.070	0.070
	LASSO	30.200	39.800	1860.200	69.800	0.302	0.355
	XGB	24.700	86.200	1813.800	75.300	0.247	0.234
	RF	37.200	52.800	1847.200	62.800	0.372	0.391
G×G 2	PIS-SVDD	90.000	51.400	1848.600	10.000	0.900	0.745
	SVDD	6.200	110.400	1789.600	93.800	0.062	0.057
	LASSO	52.800	20.600	1879.400	47.200	0.528	0.604
	XGB	38.800	93.400	1806.600	61.200	0.388	0.334
	RF	44.400	58.400	1841.600	55.600	0.444	0.439
Mixed	PIS-SVDD	99.000	29.800	1870.200	1.000	0.990	0.866
	SVDD	22.800	354.200	1545.800	77.200	0.228	0.096
	LASSO	60.800	48.600	1851.400	39.200	0.608	0.580
	XGB	28.900	96.200	1803.800	71.100	0.289	0.256
	RF	16.800	80.800	1819.200	83.200	0.168	0.167

Values are averages over 50 repetitions.

CASE 3: INTERPRETATION

- ▶ Case 3 has a sine-based nonlinear anomaly-generating structure.
- ▶ In the $C \times G$ 2 setting, PIS-SVDD achieved

$$\text{Recall} = 0.802, \quad \text{F1-score} = 0.714$$

- ▶ In the $G \times G$ 2 setting, PIS-SVDD achieved

$$\text{Recall} = 0.900, \quad \text{F1-score} = 0.745$$

- ▶ In the mixed setting, PIS-SVDD achieved the highest performance:

$$\text{Recall} = 0.990, \quad \text{F1-score} = 0.866$$

- ▶ SVDD(no screening) showed low recall and F1-score in all three settings.

Conclusion

CONCLUSION

This thesis studied one-class anomaly detection for mixed-type data when anomalies are generated by rare interaction structures.

Main contribution

- ▶ This thesis proposes PIS-SVDD, a two-stage SVDD procedure with pseudo-sample based interaction screening.
- ▶ The proposed method screens $C \times G$ and $G \times G$ interaction candidates without using anomaly labels.
- ▶ Selected interactions are added as numerical interaction features, and Stage-2 SVDD is refitted to reflect interaction-driven abnormal patterns.
- ▶ The simulation study shows that the proposed screening step improves anomaly detection performance over SVDD(no screening) across several interaction-driven anomaly settings.

LIMITATIONS AND FUTURE WORK

Limitations

- ▶ The pseudo-sample based interaction score is based on changes in the Stage-1 SVDD center distance, but theoretical properties such as sure screening consistency are not established in this thesis.
- ▶ Real data may involve many categorical levels, missingness, imbalance, and correlation among variables.

Future work

- ▶ Analyze the theoretical properties of the proposed interaction score and examine conditions for stable screening of important interaction candidates.
- ▶ Apply PIS-SVDD to real mixed-type anomaly detection data, such as medical, manufacturing, or financial data.

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